



Final Capstone Project: Complete System Automation Using Shell Script

Topic: End-to-End Automation Project

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This final capstone project is designed to combine all concepts learned in the previous labs into one complete real-world automation system. In real IT environments, automation is never done using a single small script. Instead, multiple automation tasks are connected together to form a complete system that can prepare servers, configure systems, manage users, install software, monitor health, manage logs, and maintain reliability.

This project represents a real DevOps-style automation architecture. It shows how shell scripting can be used to build a full automation pipeline that transforms a fresh server into a fully operational system automatically.

The goal of this project is not just to run commands, but to build an automated system that thinks in steps, flows logically, and works as a self-managing infrastructure unit.

In simple words, this project teaches how to build a **complete automated server management system using shell script**.



Project Objective

The objective of this project is to create one master automation script that performs multiple system-level tasks automatically. This script will simulate real-world DevOps automation used in companies for server provisioning, system setup, deployment, monitoring, and maintenance.

Project Outcomes

After completing this project, participants will be able to:

- Design full automation systems
- Combine multiple scripts into one flow
- Automate server provisioning
- Automate configuration management
- Automate application deployment
- Automate monitoring and reporting
- Build production-style automation scripts

Automation Architecture

The automation system will perform the following tasks: 1. System information collection 2. Directory structure setup 3. User creation and permission management 4. Package installation 5. Service configuration 6. Application deployment 7. Backup automation 8. Log management 9. Monitoring automation 10. Final reporting

This represents a real-world automation pipeline.

Step-by-Step Capstone Implementation

Step 1: Create Master Script

```
touch capstone.sh
```

Step 2: Open Script

```
nano capstone.sh
```

Step 3: Write Full Automation Script

```
#!/bin/bash

# =====
# CAPSTONE AUTOMATION SYSTEM
# =====

log_dir="/opt/automation/logs"
backup_dir="/opt/automation/backup"
app_dir="/opt/automation/app"
report_file="/opt/automation/final_report.txt"

mkdir -p $log_dir
mkdir -p $backup_dir
mkdir -p $app_dir

# -----
```

```

# System Info
# -----
echo "System Information" > $report_file
date >> $report_file
hostname >> $report_file
uname -a >> $report_file

# -----
# User Setup
# -----
useradd automation 2>/dev/null
echo "automation:Auto@123" | chpasswd
usermod -aG sudo automation || usermod -aG wheel automation

# -----
# Package Install
# -----
apt install -y git curl wget || yum install -y git curl wget

# -----
# Service Setup
# -----
systemctl start ssh
systemctl enable ssh

# -----
# App Deployment
# -----
touch $app_dir/app.sh

echo "#!/bin/bash" > $app_dir/app.sh
echo "echo Automation Application Running" >> $app_dir/app.sh
chmod +x $app_dir/app.sh

# -----
# Backup Task
# -----
tar -czf $backup_dir/app_backup.tar.gz $app_dir

# -----
# Monitoring Data
# -----
echo "System Health" >> $report_file
uptime >> $report_file
free -h >> $report_file
df -h >> $report_file

# -----

```

```
# Logging
# -----
log_file="$log_dir/automation.log"
echo "Automation started" > $log_file
date >> $log_file
echo "All tasks completed" >> $log_file

# -----
# Final Output
# -----
echo "CAPSTONE AUTOMATION COMPLETED"
echo "Report File: $report_file"
echo "Log File: $log_file"
```

Step 4: Permission

```
chmod +x capstone.sh
```

Step 5: Run Project

```
./capstone.sh
```

Understanding the Automation Flow

The script starts by preparing directories, then collects system information, sets up users, installs packages, configures services, deploys applications, creates backups, monitors system health, generates logs, and finally creates a report. This flow is exactly how real automation systems are designed in professional environments.

Each block represents a real operational task. Together, they form a complete automation pipeline.

Key Learning

This project teaches system thinking, automation architecture, modular scripting, and real DevOps-style automation design. It shows how shell scripting becomes an infrastructure tool.

Final Summary

- Automation is system-level thinking
 - Scripts replace manual operations
 - DevOps requires automation
 - Shell scripting builds infrastructure
 - Real systems use automation pipelines
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End of Capstone Project

This capstone represents the transformation from:

Learning shell scripting → Becoming an automation engineer

Participants now understand: - End-to-end automation - Infrastructure automation - DevOps automation architecture - Real-world scripting - System engineering mindset

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